TOUFIK SOUADIA

BACKEND ENGINEER | NODE.JS · NESTJS · TYPESCRIPT | POSTGRESQL · REDIS · ELASTICSEARCH

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PROFESSIONAL PROFILE

Backend Engineer with 5+ years of professional experience building REST APIs, ERP systems and production web applications. Core technologies include Node.js, NestJS, TypeScript, PostgreSQL, Redis and Elasticsearch, complemented by experience with Laravel, React and Docker-based delivery. Experienced across requirements analysis, API development, database design, authentication, caching, search, deployment and maintenance. PhD researcher in privacy-preserving federated learning for healthcare; winner of the 2025 national Linux competition.

TECHNICAL SKILLS

Backend & APIs: Node.js, NestJS, Express.js, TypeScript, Laravel (PHP), REST APIs, JWT/OAuth2, Odoo, ERPNext
Databases, cache & search: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, TypeORM
Infrastructure & testing: Docker, CI/CD, Linux, Nginx, Git/GitHub, Jest, PHPUnit
Frontend & desktop: React, Tailwind CSS, Inertia.js, JavaScript (ES6+), Electron
AI & research: Python, PyTorch, federated and split learning, differential privacy

PROFESSIONAL EXPERIENCE

Backend Developer | DigiBooking — 2026 - Present

Bir Mourad Raïs, Algiers · NestJS, TypeScript, PostgreSQL, TypeORM, Redis, Elasticsearch

- Designed modular REST APIs and implemented ride workflows covering requests, driver assignment, status tracking and cancellation.
- Implemented JWT/OAuth2 authentication and role-based access for passengers, drivers and administrators.
- Designed PostgreSQL schemas and migrations, optimized queries, and introduced Redis caching to reduce database load under heavy traffic.
- Built Elasticsearch full-text search and filtering with custom scoring.

Web Developer (Freelance, part-time) | Kiva Star — 2025 - 2026

Algiers · Laravel, React, PostgreSQL

- Developed a custom industrial ERP now in production, covering inventory, sales, supplier management, invoicing and reporting.

Web Developer | DevTeam — 2022 - 2025

Kouba, Algiers · PHP, Laravel, Node.js, Express.js, MySQL, PostgreSQL, React, Redis, Docker, CI/CD, Nginx

- Delivered client projects end-to-end, from requirements analysis and design through development, testing, deployment and maintenance, collaborating through Git and code review.
- Built REST APIs with Laravel and Node.js/Express, including authentication and role management, and developed React/Tailwind interfaces.
- Designed MySQL and PostgreSQL databases, introduced Redis caching for high-use APIs, and containerized applications with Docker.
- Configured CI/CD deployments to Linux servers using Nginx.

ERP Developer | ONPS - National Office of School Publications — 2021 - 2022

El Achour, Algiers · Odoo, ERPNext, Python

- Developed a custom ERP supporting the production pipeline of school publications.
- Built inventory workflows for stock movements, stock counts and traceability, alongside production-monitoring dashboards.
- Delivered the institutional website with a publication catalogue and document management; the site is used nationwide.

EARLIER EXPERIENCE Developer Intern, CERIST (Apr-May 2019): cybersecurity research project involving custom honeypots and a P2P architecture for distributed deployment and attack-data sharing. | Web Developer Intern, Glaxiom (Feb-Mar 2019): HTML, CSS and JavaScript frontend development; team Git workflows.

TOUFIK SOUADIA / BACKEND ENGINEER

Research, selected projects and academic background

RESEARCH & ACADEMIC EXPERIENCE

Health Monitoring with Federated Learning (IoMT) 2021 - Present

- PhD research on privacy-preserving healthcare monitoring, with local model training on medical devices to avoid centralizing sensitive raw data.
- Research includes multimodal fall and anomaly detection using ST-GCN and 1D-CNN on skeleton and IMU sensor data, plus adaptive sensor fusion (QCGF) designed to remain resilient to faulty sensors.
- Federated optimization with FedProx and secure aggregation with SecAgg across multiple sites.
- University teaching and supervision of final-year Bachelor's and Master's projects.

RESEARCH FOCUS IoMT · Human Activity Recognition · privacy-preserving machine learning · multimodal sensor fusion · secure aggregation

SELECTED PROJECTS

Algerian Linux System | 2025

Winner of the 2025 national Linux competition. Built a complete system from Linux From Scratch and implemented two Rust kernel drivers for USB devices.

Linux From Scratch · Rust · USB kernel drivers

Custom ERP - Desktop Edition

Customized and extended ERPNext with additional business modules. The interface covers areas including stock, production, finance, sales, purchasing and HR.

ERPNext · ERP customization · Desktop edition

Wifi Djelfa - E-commerce

Online store for computer components, laptops and accessories, with a custom PC builder, order tracking, delivery rates by wilaya, wishlist and cart; available in Arabic and French.

E-commerce · PC builder · Order tracking

Dar Jadida - Real Estate Portal

Multilingual real-estate portal with search and property discovery capabilities.

Web platform · Search · Multilingual content

EDUCATION

PhD in Computer Science (in progress) Research topic: privacy-preserving federated learning for connected healthcare.
Master's in Computer Science USTHB, Algiers | 2018 - 2020
Bachelor's in Computer Science USTHB, Algiers | 2015 - 2018

CERTIFICATIONS & LANGUAGES

NVIDIA DLI - Fundamentals of Deep Learning (2024) | Cisco Certified Network Associate (CCNA), 2022
Arabic: Native | French: Fluent | English: Intermediate